

DOÜ: From School to Home

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ABSTRACT

Before the pandemic, Quebec's education system was already in a fragile state. Now, it is in a more precarious situation. In this context, teachers, parents, and students are facing many challenges and are increasingly overloaded, stressed, and anxious. Maintaining good communication and collaboration between students, their parents, and teachers is essential. Therefore, we have developed DOÜ, a product system that acts as an intermediary between the teacher and the parent, through the child. DOÜ allows parents to strengthen their trust in teachers, children to better communicate their emotions and teachers to feel valued in their profession.

CCS CONCEPTS

• **Human-centered computing** → Interaction design; Collaborative and social computing; Collaborative and social computing design and evaluation methods; Ethnographic studies; Collaborative and social computing; Collaborative and social computing design and evaluation methods; Ethnographic studies; Interaction design; Interaction design process and methods; User centered design; • **Human-centred computing** → User centered design.

KEYWORDS

Interaction design, User-centered design, Interface design, Social interaction, Engagement, Education, Community, Communication, Gratitude

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1 SOCIOPOLITICAL CONTEXT

The classroom is a small community, a reflection of society. It is in the classroom that teachers educate the citizens of tomorrow. In this environment, children will acquire the tools to shape their personality, learn to communicate and to live in society. Therefore, teachers have a very important role to play, but they often find

themselves having to defend their interests and argue about the value of their profession. Since teaching is an occupation, not a calling.

In Canada, more specifically in Quebec, nearly 25% of new teachers leave the profession within the first seven years of their career [1]. To practice their profession, they must no longer simply get a university degree, but must now adapt to the heterogeneity of their public, use new technologies, perform administrative tasks, cooperate with their colleagues, and consider a series of ethical rules [2]. Since the beginning of the pandemic, an additional burden has been added to their already saturated work cycle, including the occasional management of online courses, sanitization of teaching tools, replacement of absent colleagues, as well as being affected by the impacts of labor shortages.

Even though the teaching profession is regularly covered in the media, the schools remain closed environments and parents do not often hear about what really goes on behind the walls. Children do not necessarily want to talk about what happens in the classroom, but for a child to develop positively, education must take place at home as well as at school. Parents are key players in children's learning. They must trust the teachers by giving them autonomy in their pedagogical approach, to evolve together towards a common goal, which is the success of the child.

2 ETHNOGRAPHIC RESEARCH PROCESS

During the ethnographic research process, qualitative research methods were used to collect data (Figure 1) and observe the behavior of various actors to develop a global view of the teaching context in Quebec [3]. It was possible to conduct interviews, complete log-books, and conduct non-participant and participant observations sessions with teachers, parents, and children [4].

2.1 Listening

We began by conducting nine interviews with parents, teachers, and principals from different socioeconomic contexts schools. These interviews provided a better understanding of the current state of teaching in Quebec, as well as the frustration that teachers are experiencing in the current context.

"It's so deeply rooted in the culture that teaching is a calling that I shouldn't ask to be paid for my overtime. In the name of calling, we are always asked for more, but when we ask for help, we are not given any. . ."
Teacher interviewed

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While these interviews led to many discussions surrounding teachers' working conditions, they also brought out important patterns that will be considered in the development of the final solution. According to many participants, the idea of building interactions based on positive feedback came up frequently and seemed to be the best way to demonstrate gratitude. Parent trust and utopia of a two-way collaboration between parents and teachers were also ideas that were mentioned in the interviews that seemed meaningful to the participants.

To explore these ideas further, five teachers agreed to complete a daily logbook, which allowed us to contextualize the concepts discussed during the interviews. The logbooks revealed that the most meaningful moments of gratitude in daily life are the least tangible, such as smiling, thanking, listening, trusting, and empowering. The logbooks also showed that the gratitude felt by the teachers can also be manifested through the success of certain teaching methods and the feeling of pride felt by the children.

2.2 Observation

We conducted two methods of immersive observations during our research phase, including three non-participants observations sessions and five participants observations sessions. To have a better understanding of the context of the project's intervention, three non-participants observations sessions were first conducted in a second grade class, a fifth grade class, and a special education class. As a result of these observations, it was possible to raise the point that teachers have overloaded schedules and often need reinforcement from different colleagues within their school. Although they sometimes get help, they still must rely on the involvement and participation of different students in the same class to be able to complete some of their daily tasks.

The first three participants observations sessions took place in the same classrooms mentioned in the previous paragraph. Highlights from these observations included students desire to participate and their trust in teachers. The need for teachers to adapt to the needs of each child and the importance of parents following up at home to facilitate classroom activities were also key elements to be remembered.

The fourth participant observation was conducted in a Quebec emergency call center. This observation strategy, used as an extreme case, pointed out more insidious behaviors that we would probably not have perceived in a usual context. This led to the idea that some individuals are not always aware of the complex process behind the delivery of a service, which influences their perception of this service.

Finally, the last participant observation was conducted by the four members of our team in a Pound class at a yoga center in Quebec City. Following this course, it was possible to raise different findings (Figure 1). More precisely, external factors have a great importance on learning, the group effect influences the service delivery, the level of knowledge influences learning and autonomy as well as the importance of the quality of the pedagogical method on learning.

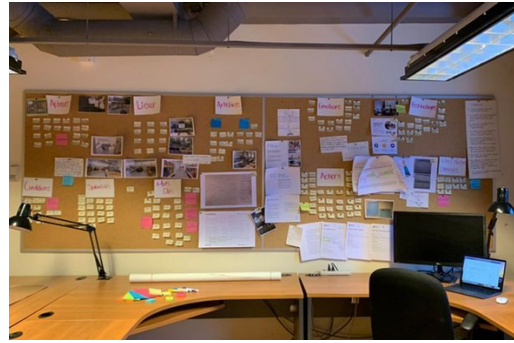


Figure 1: Analysis of the results generated during the ethnographic research

3 DESIGN OPPORTUNITY

During the ethnographic research process, teachers regularly mentioned the extent of parent's involvement in their children's education, as well as the trust and autonomy they allow them in the application of their teaching methods. They also repeatedly mentioned the importance placed on student's perception of their personal success. According to the teachers, feeling valued, supported, and sustained, when implementing the methods used for the child's success, are signs of gratitude in themselves. It is the "slap on the back" they need to keep going. These intangible signs of recognition are far more meaningful and allow them to build a sense of resilience, being more nurtured and fulfilled at work [5].

As a result of these findings, our team then focused on the importance of opening the walls of the school by facilitating communication between the parent and teacher, as well as focusing this communication on positive notes, so that they could collaborate towards the child's success (Figure 2). Strengthening this relationship as a design opportunity seems, at this point, the ideal way to reach the community with an awareness of the magnitude of teachers' daily tasks. The importance of designing communicative places or spaces based on human relationships rather than machines has become part of our daily lives and allows us to address much larger contexts [6]. By raising awareness to people individually, we then hope to incite an intrinsic change in the perception of the job in the longer term and on a larger scale.

4 RELATED WORKS

The development of products in the context of human-computer interaction aimed to reconnect parents with their children inside the school walls is not new. Applications or software are currently part of the daily life of teachers, allowing them to communicate freely with parents and children to promote academic monitoring outside the classroom. Among the most popular apps, ClassDojo directly targets the reconnection of teachers with students and parents to create incredible classroom communities [7]. The Seesaw app also regularly used by teachers, allows them to visualize children's learning process, while promoting parent engagement [8]. Other digital tools present in the school environment also promote better communication between the two spheres of education of the child, such as the digital portfolio financed by the *Ministère de l'Éducation*

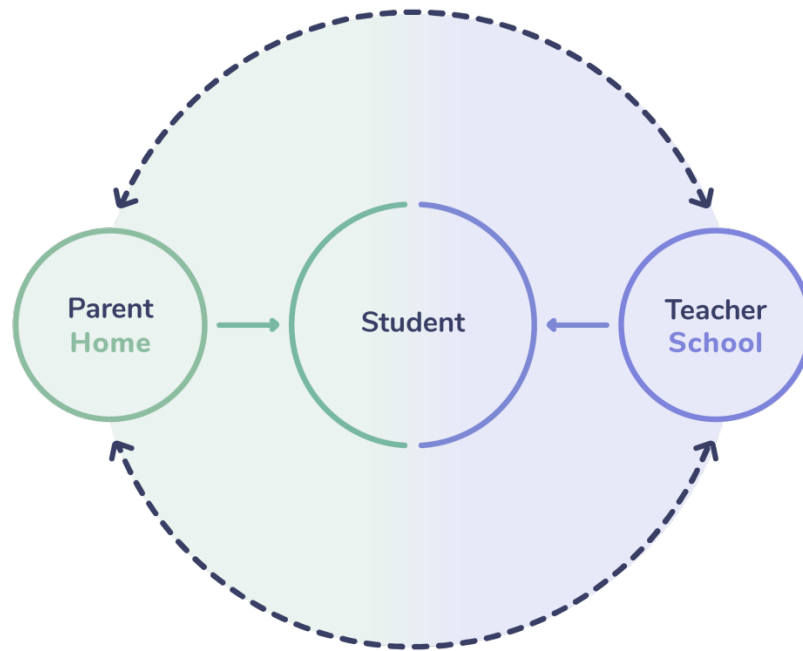


Figure 2: Relationship between key actors of the school system

et de l'Enseignement supérieur [9], the Google Suite [10] or the Mozaïk portal [11].

Although effective, these tools sometimes tend to overload teachers in their daily tasks. Some of these applications allow parents to share the information they want, without a filter, and to communicate with teachers when they want to. The teachers we met also raised the lack of emotional involvement of the children in the operation of these educational tools. There is also a temporal and emotional gap between what happens in the classroom and when the information is sent, particularly due to the lack of time from teachers. This study of existing tools in the school context in Quebec has identified some key elements that are important to consider for the design of the final proposal.

5 DOÛ – FROM SCHOOL TO HOME

Our final proposal DOÛ (Figure 3) sits at the heart of the interactions between teachers and parents, through the child, who becomes the intermediary between the two. It is a physical product inspired by the tin can telephone, accompanied by an application available on mobile and computer, on which the messages entrusted by the students will be collected. A small gauge installed with a plug-in on the interactive whiteboard will allow seeing in real-time all the gratitude received. This product system is part of a sensory experience that allows the different actors to promote well-being together in each environment, the classroom.

5.1 The child confides to DOÛ

Under the direction of the teacher, the child goes to a corner of the classroom and confides to DOÛ, with the aim to highlight his or her good deed, on the spot. The device recognizes the student's fingerprint and associates it directly with his profile on the application. The voice message, of a maximum of 30 seconds, allows the child to learn how to express and communicate his emotions.

5.2 The teacher receives the message from the student and forwards it to the parent

The message communicated by the child is then sent to the application and received by the teacher. The teacher will then be able to listen to the message and decide if he want to forward it to the parent.

5.3 The parent receives the message

The parent receives the message on the application, which allows them to follow up at home with their child. They can then respond to the teacher's message by showing their appreciation using a bank of predetermined stickers.

5.4 Gratitude is visible

All the positive messages received will make DOÛ fill up with gratitude. The more messages of good deeds and follow-ups DOÛ receives, the more intense its light will be. This intensity will be



Figure 3: Sequence of use of the DOÛ product system, this image has been designed using resources from ©Freepik.com



Figure 4: Validation session with an elementary school student

visible on the device, the application and through the plug-in on the interactive whiteboard.

6 PARTICIPATORY EVALUATION METHOD

Given the current progress of the project, the participatory evaluation method in the form of an interview was preferred [12]. The solution was therefore presented in the form of a conceptual argument to four elementary school teachers to obtain their impressions. This approach helped to validate the sequence of use and to better define the context, to improve the overall user experience. A preliminary ergonomic evaluation was also conducted with a child to validate the desired size and shape for the next steps of the development of the solution (Figure 4). These iterative evaluation methods

allowed us to develop the proposal revealed above, so that it would fit properly into its context.

7 CONCLUSION

By using an ethnographic research approach and working in tandem with several teachers in advance of the project, the product system DOÛ demonstrates an accurate understanding of the context in which it is embedded. Gratitude expands ways of thinking and helps build more sustainable social relationships, resulting in increased teacher fulfillment at work [13]. In the current climate where teachers are leaving the profession at a catastrophic rate because they are stressed and overworked, a more focused attention on gratitude, trust, and autonomy would reverse this trend [5].

Although some validation has already been done, more evaluation will be needed for the continuation of the project. First, a more accurate prototype of the device will be needed to test the final proposal. This step can be done in conjunction with the validation of the fingerprint recognition technology. This technology is based on physical biometric aspects, with the aim to associate each student with the right profile on the platform [14]. Finally, it will be necessary to validate the sequence of use of the product system in its context of use, to arrive at a coherent proposal adapted to the current educational system.

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